

Unraveling the Story of the Apollo Lunar Thin Section 76539-10-r1

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1 Introduction

This lunar rock 76539 is a small aphanitic basalt found in the rake sample collected at station 6, Apollo 17. The word *aphanitic* means fine-grained. Seen from the following image of sample 76539, this lunar rock is so fine-grained that individual mineral crystals cannot be seen with the naked eye. An aphanitic texture forms when magma cools rapidly, preventing crystals from growing large.



Figure 1: Apollo lunar sample 76539 from Apollo 17. Credit: Virtual Microscope

2 Petrographic Observations

Paired rotational series of PPL and XPL images of sample 76539-10-r1 at 5° intervals was acquired from Virtual Microscope [1] and subsequently aligned to ensure a consistent field of view.



Figure 2: Sample 76539-10-r1 in PPL (top) and XPL (bottom). Field of view 2.0 mm

2.1 Skeletal grains

In PPL, as seen in Figure 2, several distinct skeletal grains are present:

Elongate chain /swallowtail crystal. The most conspicuous grain is a long, narrow crystal crossing the upper-central part of the field on a diagonal. It is not a solid blade: it shows a segmented, internally hollow structure and a forked (swallowtail) termination.

Hopper (hollow-box) crystals. At least one well-defined hollow, box-shaped cross-section sits on the right side of the field, and additional hollow frames appear in the lower-left cluster. These are hopper sections: in PPL, they read as bright rectangular outlines enclosing dark centers.

Open lattice /grid frameworks. The lower-left quadrant contains several crystals built from parallel and cross-connected bars, producing a grid- or ladder-like appearance. This is the skeletal end of the morphology spectrum, intermediate between hopper and fully dendritic growth.

In XPL, the skeletal grains display a range of moderate-to-high interference colors, turquoise and blue, gold and yellow, tan, and patches of lavender, set against the extinct matrix.

Interpretation:

The relatively vivid colors, combined with the absence of obvious cleavage on the crystal outlines and the hopper/skeletal habit, are consistent with olivine rather than a lower-birefringence phase. Confident phase assignment for any individual grain would require compositional measurement, but the assemblage as a whole matches the published skeletal-olivine-plus-glass description of 76539 [2, 3]. Elongation with hollow segments and bifurcating tips is diagnostic of growth at high undercooling, where extension along a fast-growth direction outpaces infilling of the crystal faces. Hopper sections infer that the crystal edges and corners are developed while the interior remains an open cavity, subsequently occupied by glass [2–5].

2.2 Needle-like opaque grains

In a lunar thin section, groundmass is the fine-grained assemblage of materials and /or glass that surrounds the larger crystals; whereas phenocrysts are the large, early-formed crystals surrounded by a fine-grained groundmass. In both PPL and XPL, the bulk of the field is dark. In XPL, this matrix stays extinct across the field, behaving isotropically.

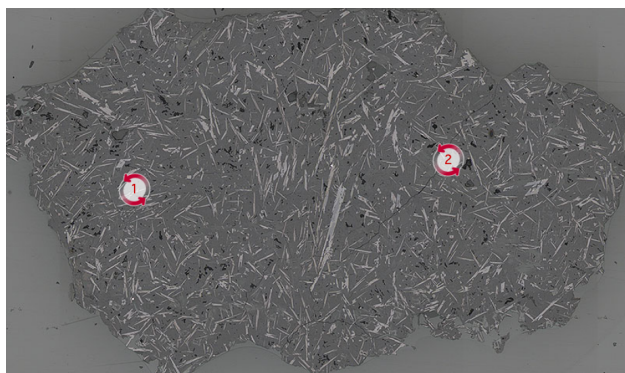


Figure 3: Sample 76539 under reflected light

As seen from the relected light image of lunar rock 76539 in Figure 3, the needle-like opaque grains appear bright and highly reflective; whereas they remain black in both PPL and XPL at all stage rotations, exhibiting skeletal to lath-shaped morphologies characteristic of ilmenite.

2.3 Groundmass

Glass is amorphous, so it's optically isotropic, which means it stays extinct (black) in XPL at every stage rotation with no birefringence. Normally, in PPL, ordinary glass transmits light. Clear glass looks colorless /clear. The groundmass appears black in PPL and black in XPL, which means the diagnostic meaning is simply that it is not transmmitting light at all. It's opaque in transmitted light. Exinction in XPL means isotropic or extinct. Opacity in PPL means that no light gets through. Black in both therefore has two main candidates: (1) opaque minerals, ilmenite and other Fe-Ti oxides, metal, or sulfides, or (2) opaque glass which is glass so dark, iron-rich, and loaded with submicroscopic quench crystallites that it does not transmit visible light. Thus, the groundmass is mostly opaque glass, in agreement with the vitrophyric character reported for 76539 [2].

3 Interpretation

The textures recorded in 76539-10-r1 form an internally consistent quench assemblage. Skeletal and hopper olivine develops when a melt is cooled well below the olivine liquidus, so that crystal growth becomes diffusion-limited: edges and corners projecting into fresh melt grow rapidly while the faces, bordered by chemically depleted boundary layers, lag behind. The result is preferential edge growth and starved interiors — the hollow boxes, chains, and lattices seen here. The progression visible within this one field, from hopper boxes through open grid frameworks toward branching forms, mirrors the morphological sequence expected with increasing undercooling.

The glassy, optically isotropic groundmass reinforces this reading. It indicates that, after the olivine skeletons had grown, the residual melt was quenched too quickly to crystallize, freezing in as glass. Taken together with the elongate swallowtail crystal and the hopper sections, the field documents a single, rapid cooling history — exactly what is expected for a thin, surface-chilled high-Ti basalt such as 76539.

The high-Ti bulk composition of the sample is also consistent with the petrographic context: titanium in these basalts is carried chiefly by ilmenite, reported in 76539 as fine needles best seen in reflected light [3].

4 Conclusion

- 1) Thin section 76539-11-r1 preserves a population of skeletal and hopper olivine crystals within an isotropic glassy groundmass.
- 2) The paired rotational sequence allows the same crystals to be read both morphologically and optically; the moderate-to-high interference colors support an olivine identification for the skeletal grains.
- 3) The assemblage is a textbook quench texture, recording olivine growth under strong undercooling followed by rapid quenching of the residual melt to glass.

- 4) These observations are consistent with the established characterization of 76539 as a small, rapidly cooled high-Ti vitrophyric mare basalt from Apollo 17.

References

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